



JKBOSE 2026

CLASS 11TH

SEQUENCES AND SERIES

MATHS

LECTURE-01

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TOPICS TO BE COVERED

1 What is Sequence and Series?

2 Exercise Discussion 8.1

3 Related Problems

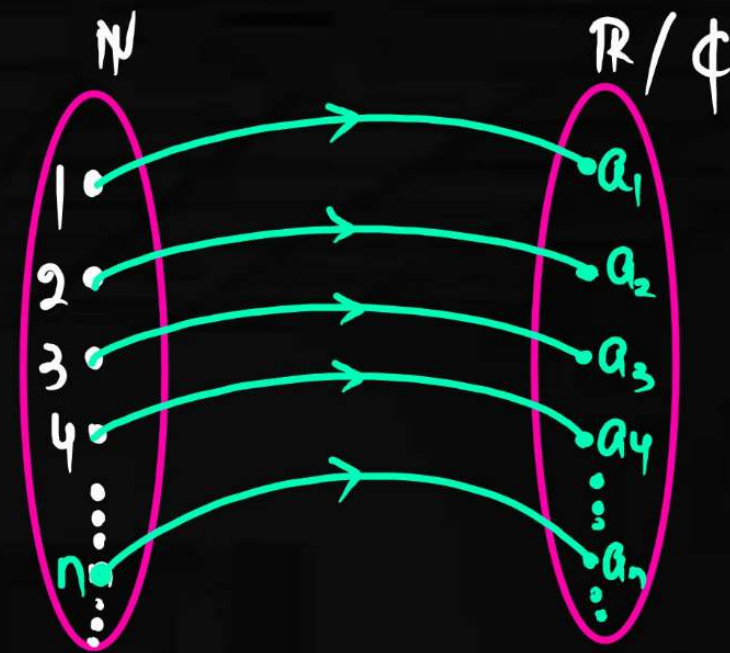
INTRODUCTION

* Any list of Numbers is known as a sequence.

2, 4, -11, -15, 16, 18, ...
↓ ↓ ↓ ↓ ↓ ↓
 a_1 a_2 a_3 a_4 a_5 a_6

* Members of the Sequence are known as the terms of the sequence.

* Generally denoted by T_n or a_n .
(n^{th} term of the sequence)



Definition: A sequence (infinite) is a function whose domain is the set of Natural numbers.

* $f: N \rightarrow R$ (Real sequence) ✓

* $f: N \rightarrow C$ (Complex sequence)

* 2, 4, 6, 8, ... (Sequence with a pattern)
↓ ↓ ↓ ↓
 a_1 a_2 a_3 a_4

Progression

* A sequence with a pattern is known as a progression.

* 2, 4, 6, 8, 10, 12, ... (Progression)
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $2 \times 1 \quad 2 \times 2 \quad 2 \times 3 \quad 2 \times 4$

Here $a_n = 2n \quad \forall n \in \mathbb{N}$

* All the terms of the progression can be obtained from a General formula.

* $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots (a_n = \frac{1}{n})$

* 3, 7, 11, 15, 19, ... (Arithmetic Progression)
 $a_n = a + (n-1)d$
 $S_n = \frac{n}{2} \{2a + (n-1)d\}$

Types of Progression

Arithmetic Progression
(A.P.)

Geometric Progression
(G.P.)

Harmonic Progression
(H.P.)

Arithmetico-Geometric Progression
(A.G.P.)

EXAMPLE

Write the first three terms in each of the following sequences defined by the following:

(i) $a_n = 2n + 5,$

(ii) $a_n = \frac{n-3}{4}$

(i)(Sol) Here $a_1 = 2 \times 1 + 5 = 7.$

$$a_2 = 2 \times 2 + 5 = 9$$

$$a_3 = 2 \times 3 + 5 = 11.$$

(ii)(Sol) Here, $a_1 = -\frac{1}{2}, a_2 = -\frac{1}{4}, a_3 = 0.$

EXAMPLE

A sequence may be defined as a

(a) ~~X~~ Relation, whose range $\subseteq \mathbb{N}$ (natural number)

~~(b)~~ Function whose range $\subseteq \mathbb{N}$

~~(c)~~ Function whose domain $\subseteq \mathbb{N}$

~~(d)~~ None of the above

QUESTION

The 18th term of the sequence defined by

$$a_n = \frac{n(n-3)}{n+4}$$

1 $\frac{140}{11}$ ✗

2 $\frac{135}{11}$ ✓

3 $\frac{135}{12}$ ✗

4 $\frac{140}{16}$ ✗

$$a_{18} = \frac{9 \times 15}{22} \Rightarrow a_{18} = \frac{135}{11}$$

* 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...
(Fibonacci sequence)
Sequence whose each term is the sum of the two preceding terms.

EXERCISE – 8.1

QUESTION

Write the first five terms of each of the sequences in Exercises 1 to 6 whose n^{th} terms are:

$$a_n = n(n + 2)$$

$$(Sol) \quad a_1 = 1 \times 3 = 3$$

$$a_2 = 2 \times 4 = 8$$

$$a_3 = 3 \times 5 = 15$$

$$a_4 = 4 \times 6 = 24$$

$$a_5 = 5 \times 7 = 35$$

QUESTION

Write the first five terms of each of the sequences in Exercises 1 to 6 whose n^{th} terms are:

$$a_n = \frac{n}{n+1} \neq \text{H/W}$$

QUESTION

Write the first five terms of each of the sequences in Exercises 1 to 6 whose n^{th} terms are:

$$a_n = 2^n$$

$$2^1, 2^2, 2^3, 2^4, 2^5$$

#

QUESTION

Write the first five terms of each of the sequences in Exercises 1 to 6 whose n^{th} terms are:

$$a_n = \frac{2n - 3}{6}$$

HP/W.

QUESTION

Write the first five terms of each of the sequences in Exercises 1 to 6 whose n^{th} terms are:

$$a_n = (-1)^{n-1} \underline{\underline{5^{n+1}}}$$

QUESTION

Write the first five terms of each of the sequences in Exercises 1 to 6 whose n^{th} terms are:

$$a_n = n \frac{n^2 + 5}{4}$$

QUESTION

Write the first five terms of each of the sequences in Exercises 1 to 6 whose n^{th} terms are:

$$a_n = 4n - 3; a_{17}, a_{24}$$

$$\begin{aligned} a_{17} &= 4 \times 17 - 3 \\ &= 65 \end{aligned}$$

$$\begin{aligned} a_{24} &= 4 \times 24 - 3 \\ &= 193. \end{aligned}$$

QUESTION

Write the first five terms of each of the sequences in Exercises 1 to 6 whose n^{th} terms are:

$$a_n = \frac{n^2}{2^n}; a_7$$

h/w.

$$\frac{7^2}{2^7} = \frac{49}{128}$$

QUESTION

Write the first five terms of each of the sequences in Exercises 1 to 6 whose n^{th} terms are:

$$a_n = (-1)^{n-1} n^3; a_9$$

$$\# \text{ (Q13) (Sol) } a_1 = a_2 = 2.$$

$$a_n = a_{n-1} - 1, n > 2$$

$$a_3 = a_2 - 1 = 1$$

$$a_4 = a_3 - 1 = 0$$

$$a_5 = a_4 - 1 = -1.$$

QUESTION



Write the first five terms of each of the sequences in Exercises 1 to 6 whose n^{th} terms are:

$$a_n = \frac{n(n-2)}{n+3}; a_{20}$$

$$a_{20} = \frac{20 \times 18}{23} = \frac{360}{23}$$

$$\frac{a_{n+1}}{a_n} \quad n=1,2,3,4,5.$$

(Q14)

* (Sol) Given that $a_1 = a_2 = 1$ & $a_n = a_{n-1} + a_{n-2}$.

Here the Fibonacci Sequence is:

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, ...

$$\text{So, } \frac{a_2}{a_1} = 1, \frac{a_3}{a_2} = 2, \frac{a_4}{a_3} = \frac{3}{2}, \frac{a_5}{a_4} = \frac{5}{3}, \frac{a_6}{a_5} = \frac{8}{5}.$$

EXAMPLE



let $\{a_n\}$ be the sequence such that $a_1 = 1$ and $3a_{n+1} - 3a_n = 1 \forall n \geq 1$. Then find the value of a_{2002}

(Sol) Given that $3a_{n+1} - 3a_n = 1$ & $a_1 = 1$.

$$\Rightarrow 3(a_{n+1} - a_n) = 1.$$

$$\Rightarrow a_{n+1} - a_n = \frac{1}{3} \quad \text{--- (1)}$$

Put $n = 1, 2, 3, 4, 5 \dots 2001$, we get

$$a_2 - a_1 = \frac{1}{3}$$

$$a_3 - a_2 = \frac{1}{3}$$

$$a_4 - a_3 = \frac{1}{3}$$

$$a_5 - a_4 = \frac{1}{3}$$

$$a_6 - a_5 = \frac{1}{3}$$

$\vdots$

$$a_{2001} - a_{2000} = \frac{1}{3}$$

$$a_{2002} - a_{2001} = \frac{1}{3}$$

Adding the above equations, we get

$$\begin{aligned} & (a_2 - a_1) + (a_3 - a_2) + (a_4 - a_3) + (a_5 - a_4) + (a_6 - a_5) \\ & + \dots + (a_{2001} - a_{2000}) + (a_{2002} - a_{2001}) = \frac{1}{3} \times 2001. \end{aligned}$$

$$\Rightarrow a_{2002} - a_1 = \frac{2001}{3}$$

$$\Rightarrow a_{2002} = \frac{2001}{3} + a_1$$

$$\Rightarrow a_{2002} = \frac{2001}{3} + 1$$

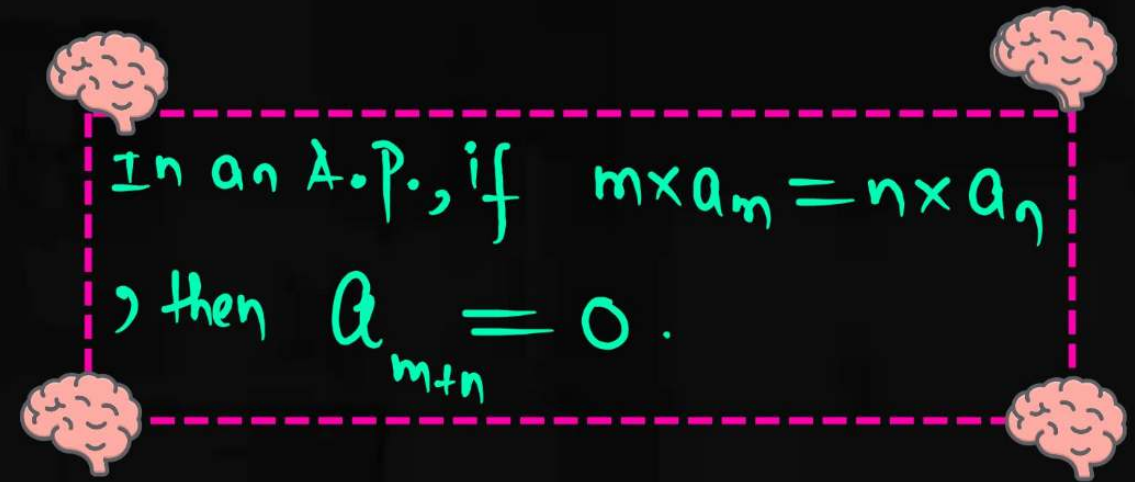
$$\Rightarrow a_{2002} = \frac{\cancel{2004}^{668}}{\cancel{3}_1}$$

$$\Rightarrow \boxed{a_{2002} = 668.} \text{ Hence the result.}$$

QUESTION (NCERT EXAMPLAR)

If 9 times the 9th term of an A.P. is equal to 13 times the 13th term, then the 22nd term of the A.P. is

- | | | |
|----------|-------|---|
| 1 | 0 ✓ | (Sol) Given $9 \times a_9 = 13 \times a_{13}$ |
| 2 | 22 ✗ | $\Rightarrow 9(a + 8d) = 13(a + 12d)$ |
| 3 | 220 ✗ | $\Rightarrow 9a + 72d = 13a + 156d$ |
| 4 | 198 ✗ | $\Rightarrow 4a + 84d = 0$ |
| | | $\Rightarrow 4(a + 21d) = 0$ |
| | | $\Rightarrow a + 21d = 0$ |
| | | $\Rightarrow a_{22} = 0$ |



QUESTION (NCERT EXAMPLAR)

If the sum of n terms of an A.P. is given by $S_n = 3n + 2n^2$ then the common difference of the A.P. is

- | | | |
|----------|----------------|---|
| 1 | 3 X | (Sol) Here $S_n = 3n + 2n^2$. |
| 2 | 2 X | $\Rightarrow S_1 = 3 \times 1 + 2 \times 1 = 5 = a_1$ |
| 3 | 6 X | $S_2 = 3 \times 2 + 2 \times 4 = 14$ |
| 4 | 4 ✓ | $\Rightarrow a_2 + a_1 = 14$ |
| 5 | 6 ✓ | $\Rightarrow a_2 = 14 - a_1 = 14 - 5 = 9$ |
| 6 | 4 ✓ | So, $d = a_2 - a_1 = 9 - 5$ |
| 7 | 4 ✓ | $\Rightarrow d = 4$ |

QUESTION (NCERT EXAMPLAR)

Let S_n denote the sum of the first n terms of an A.P. If $S_{2n} = 3S_n$ then $S_{3n} : S_n$ is equal to

1 $\times$ 4

2 $\checkmark$ 6

3 $\times$ 8

4 $\times$ 10

(Sol) Given that
 $S_{2n} = 3 \times S_n$

$$\Rightarrow \frac{2n}{2} \{2a + (2n-1)d\} = 3 \times \frac{n}{2} \{2a + (n-1)d\}$$

$$\Rightarrow 2(2a + 2nd - d) = 6a + 3nd - 3d$$

$$\Rightarrow 4a + 4nd - 2d = 6a + 3nd - 3d$$

$$\Rightarrow 4nd - 2d - 3nd + 3d = 6a - 4a$$

$$\Rightarrow 2a = nd + d$$

$$\Rightarrow 2a = (n+1)d$$

$$\text{So, } \frac{S_{3n}}{S_n} = \frac{\frac{3n}{2} \{2a + (3n-1)d\}}{\frac{n}{2} \{2a + (n-1)d\}}$$

$$\Rightarrow \frac{S_{3n}}{S_n} = \frac{3(nd + d + 3nd - d)}{(nd + d + nd - d)}$$

$$\Rightarrow \frac{S_{3n}}{S_n} = \frac{12nd}{4nd} = 6$$

QUESTION

Four numbers are in A.P. The **sum of first and last term is 8** and the **product of both middle terms is 15**. The least number of the sequence is :

- 1 $\times$ (Sol) let the given numbers be $a, a+d, a+2d, a+3d$.
- 2 $\times$ Therefore $a + a+3d = 8$.
 $\Rightarrow 2a+3d = 8$.
- 3 $\times$ $\Rightarrow a = \left(\frac{8-3d}{2}\right)$ — (1)
- 4 $\checkmark$ Also, $(a+2d)(a+d) = 15$ — (2)
using (1) in (2), we get;

$$\left(\frac{8-3d}{2} + 2d\right)\left(\frac{8-3d}{2} + d\right) = 15.$$

$$\Rightarrow \left(\frac{8+d}{2}\right)\left(\frac{8-d}{2}\right) = 15.$$

$$\Rightarrow (8+d)(8-d) = 60$$

$$\Rightarrow 8^2 - d^2 = 60$$

$$\Rightarrow d^2 = 64 - 60 = 4.$$

$$\Rightarrow d = 2, -2.$$

$$\text{So, } a = \frac{8-6}{2}, \frac{8+6}{2}$$

$$\Rightarrow a = 1, 7.$$

Therefore the require A.P.'s

1, 3, 5, 7

or 7, 5, 3, 1.

Alternative Method:

let the given numbers be:

$$a-3d, a-d, a+d, a+3d \text{ (Common difference} = 2d)$$

$$\text{So, } a-3d + a+3d = 8$$

$$\Rightarrow 2a = 8$$

$$\Rightarrow a = 4$$

$$\text{Also } (a-d)(a+d) = 15$$

$$\Rightarrow (4-d)(4+d) = 15$$

$$\Rightarrow 16-d^2 = 15$$

$$\Rightarrow d^2 = 1$$

$$\Rightarrow d = 1, -1.$$

So, Required A.P.'s are:

$$1, 3, 5, 7$$

$$\text{or } 7, 5, 3, 1.$$

For 3-Terms $\rightarrow a-d, a, a+d$

For 4-Terms $\rightarrow a-3d, a-d, a+d, a+3d$

For 5-Terms: $a-2d, a-d, a, a+d, a+2d.$

Thank
You